

CSC-648-01-SU26

Project: Student Center

Team #4: BigBacks

Version History

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Team Members

Wen Chien Yen	Team Lead / Scrum Master	wyen@sfsu.edu
Oscar Wang	Frontend Lead / Technical Writer	owang1@sfsu.edu
Banghao Yuan	Software Architect	Byuan1@sfsu.edu
Sean Wang	Database Administrator	swang29@sfsu.edu
Enoch Lin	Github Master	elin5@sfsu.edu
Kashif Zada	Backend Lead	kzada@sfsu.edu

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Executive Summary

Student Center - with Significant Improvements

Our project idea is a Student Center web application. The goal of this project is to significantly improve upon the student academic planning experience by making common Student Center functions more immediately accessible and easier to understand, while also providing useful features that were otherwise not provided. Instead of redesigning Canvas, which focuses mainly on assignments, grades, and course materials, our project focuses on the administrative and planning aspect of a student's academics. This application will help students view and organize their class schedule, locate class buildings on a campus map, track degree requirements, and receive guidance when planning future courses. This system is especially helpful for new students, transfer students, international students, and students who are close to graduation. By improving the user experience of a student center, our goal is to reduce confusion during registration and help students make better academic decisions.

List of tentative unique features

In this section we will overview why the tentative unique features are unique and why competitors might not be implementing them.

Feature 1: Class Schedule Integration with Campus Map

This feature allows students to view their class schedule, and when prompted, they can be shown the location of the buildings where their classes take place on an integrated campus map. This is especially useful for new students, since they would otherwise need help from other sources or other students to locate these buildings.

Why is it unique? Many student center systems show a student's class schedule, and many university websites provide a campus map, but these tools are usually separated. Our feature combines the class schedule and campus map into one connected experience. Instead of manually searching for each building, students can directly connect their schedule to the physical campus layout.

Competitors in the same domain, such as traditional student portals, usually focus on registration, tuition, grades, and basic schedule information. Campus maps are often provided as separate web pages and are not deeply connected to a student's personal schedule. We believe many existing systems do not implement this because student information systems and campus map systems are often managed separately by different university departments.

Feature 2: Degree Requirement Tree/Roadmap

This feature displays a proper roadmap for students to understand what requirements they will need for graduation. When students select or plan courses, the system will show how those courses relate to their remaining requirements through a more user-friendly display. Requirements can be separated into categories such as major required courses, major elective courses, general education courses, and remaining total credits.

We believe this feature is unique since many students often find class registration already difficult and frustrating because degree requirement and prerequisite information is often spread across degree audit pages, department websites, advising documents, and catalog requirements. Our feature presents this information in a more student-friendly way by introducing a specific format that connects course planning directly into a roadmap for graduation.

Existing competition student center systems may already provide degree audit tools, but they are often difficult to read and not designed around course planning. They may show completed and missing requirements, but they do not always clearly guide students while they are actively selecting future courses. We believe competitors may not implement this in a more user-friendly way because degree requirement rules can be complex, especially with different catalog years, transfer credits, substitutions, and major-specific exceptions.

Feature 3: Advanced Courses and Graduate Planning Suggestions

This feature helps students who are close to completing their undergraduate degree plan ahead. If a student has remaining space in their semester schedule, the system can suggest advanced courses that may support future academic goals, such as graduate school preparation. The system would not guarantee that a course counts toward graduate credit, but it would recommend possible courses for the student to discuss with an academic advisor.

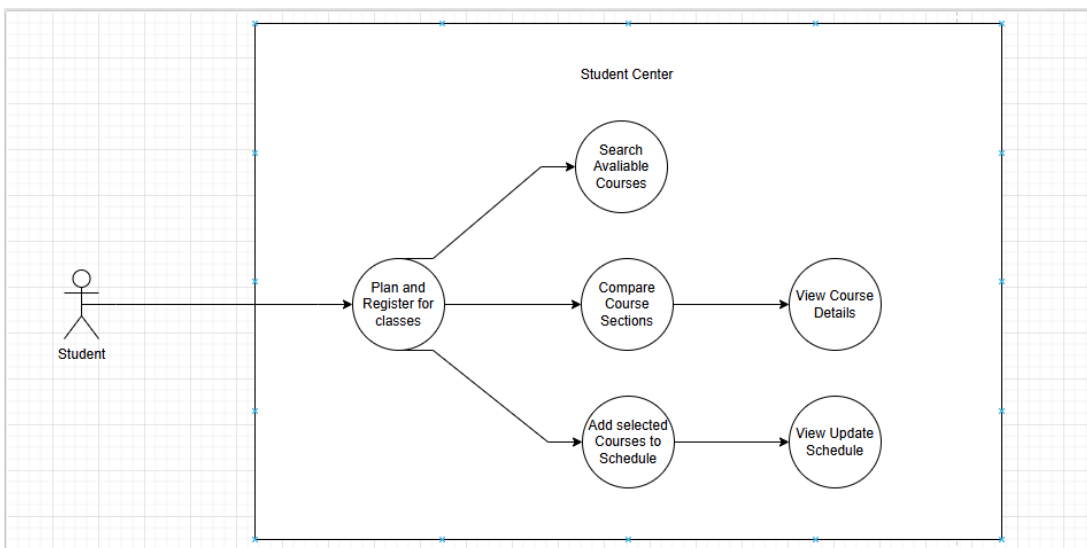
This is unique since most student center systems focus on helping students finish their current degree requirements. Our feature goes one step further by helping students plan ahead about future academic goals while also helping save time. It is designed for students who are close to graduation and want to use their remaining time more strategically.

Competitors may avoid this type of feature because graduate credit policies can be different across universities, departments, and programs. Since approval often depends on university policy and advisor review, many systems may not want to make automatic recommendations. Our approach is different because we frame the feature as planning guidance, not as an official approval system. The system can help students identify possible opportunities while still encouraging them to confirm the details with an advisor.

Use Cases

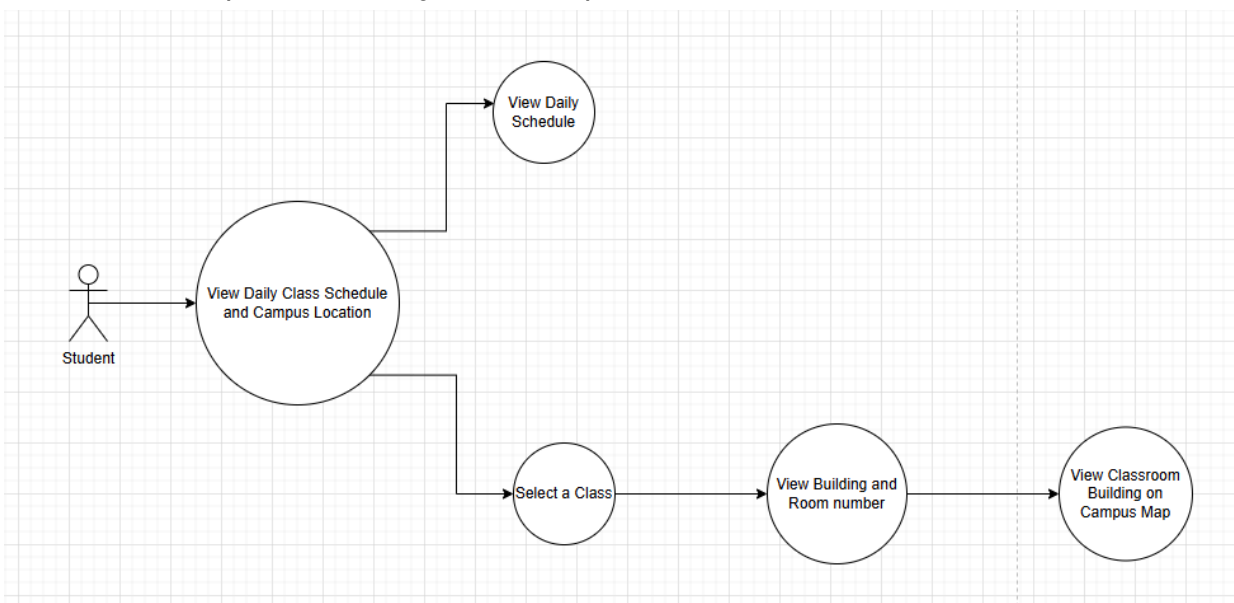
Use Case 1

- Actor
 - Student
- Assumptions
 - The Student has access to Our Student Center
 - The student has an active student account
 - Course and section information is available in the system
- Use Case
 - A student is preparing for the next semester and feels unsure about which class to take. The student wants to register for classes without switching between multiple confusing pages. The student opens our Student Center, and searches for available courses based on the academic need, schedule preferences and degree requirements.
 - The student compares course options, checks available sections and reviews important details such as course title, meeting days, time, instructor, classroom, and building. After deciding which classes fit their plan, the student adds selected courses to their schedule. The student feels more confident because their class schedule is shown clearly in one place.
- Benefits for Student
 - Helps the student choose classes more easily
 - Reduces confusion during registration
 - Allows the student to understand how selected courses fit into their academic plan.
 - Saves time by showing course and schedule information in one centralized system.



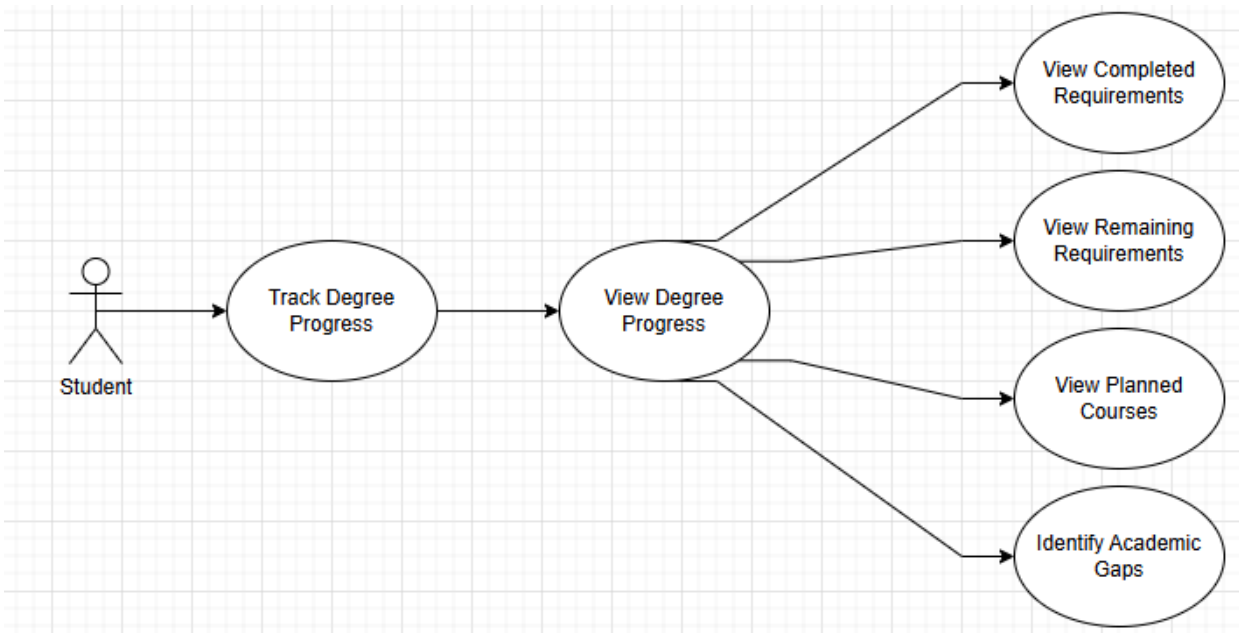
Use Case 2

- Actor
 - Student
- Assumptions
 - The student is enrolled in at least one class.
 - The student's schedule and classroom information are available in the system
 - Campus building location data exists in the system
- Use Case
 - A student is on campus and needs to attend class on time. The student may not know where a building is located, especially if they are new to campus or have classes in unfamiliar buildings. The student opens our student center or view their class schedule for the day.
 - The system shows the student the course name, meeting time, building and room number. When the student selects a class, the campus map shows where the classroom building is located. The student feels less anxious because they can quickly understand both when the class starts and where they need to go
- Benefits for Student
 - Helps students arrive at class on time.
 - Reduces stress for new students unfamiliar with campus
 - Combines schedule and location information in one place
 - Improves the daily student experience.



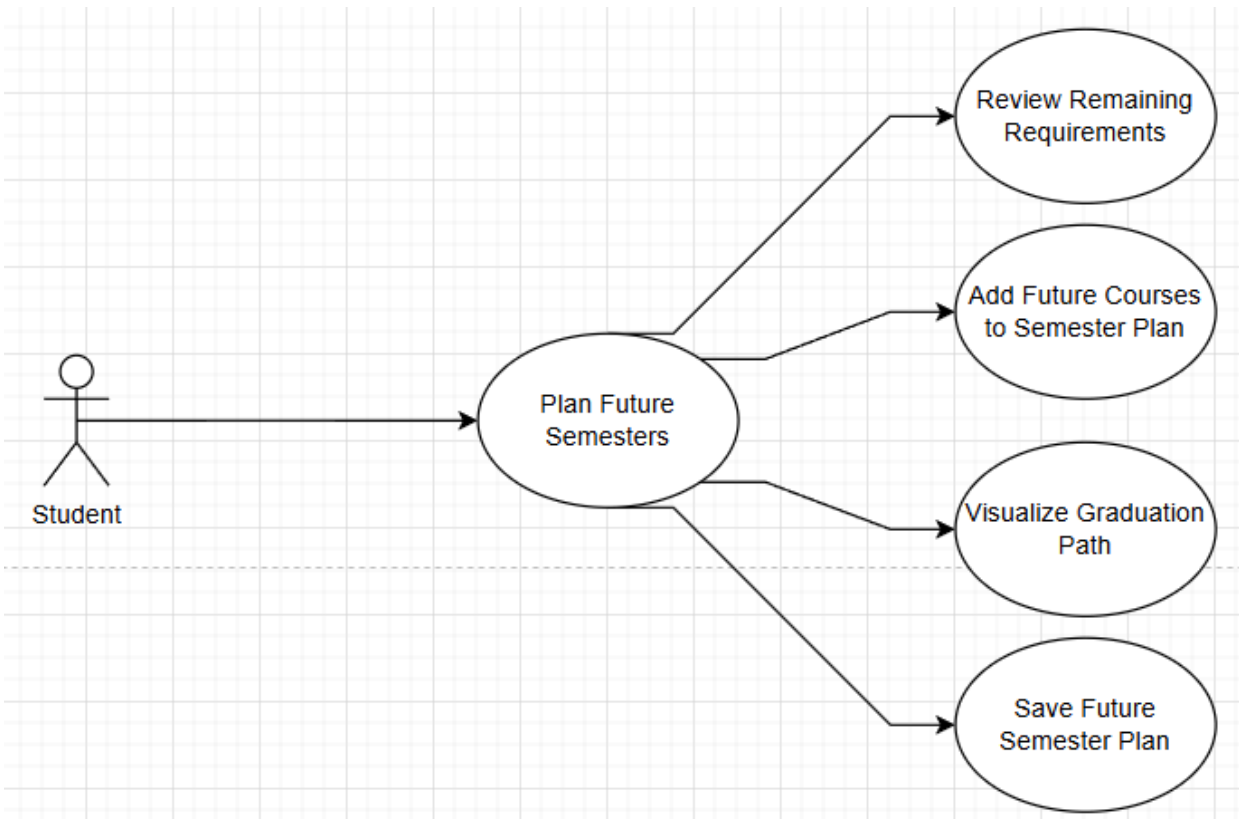
Use Case 3

- Actor
 - Student
- Assumptions
 - The student has declared a major or academic program.
 - Degree requirement data exists in the system.
 - The student's completed, enrolled, and planned courses are available.
- Use Case
 - A student wants to understand how close they are to graduation. The student may feel uncertain about where they are taking the right classes or missing important requirements. Instead of manually comparing transcripts with major requirements, the student opens our Student Center to view degree progress.
 - The System shows completed requirements, remaining requirements, planned courses, and possible gaps in the student's academic path. The student can see which major, elective, and general education requirements still need attention. This gives the student a clearer understanding of their academic progress and helps them make better planning decisions.
- Benefits for Student
 - Helps the student understand graduation progress
 - Reduce uncertainty about remaining requirements
 - Supports better long term academic planning.
 - Helps students avoid taking unnecessary or incorrect courses.



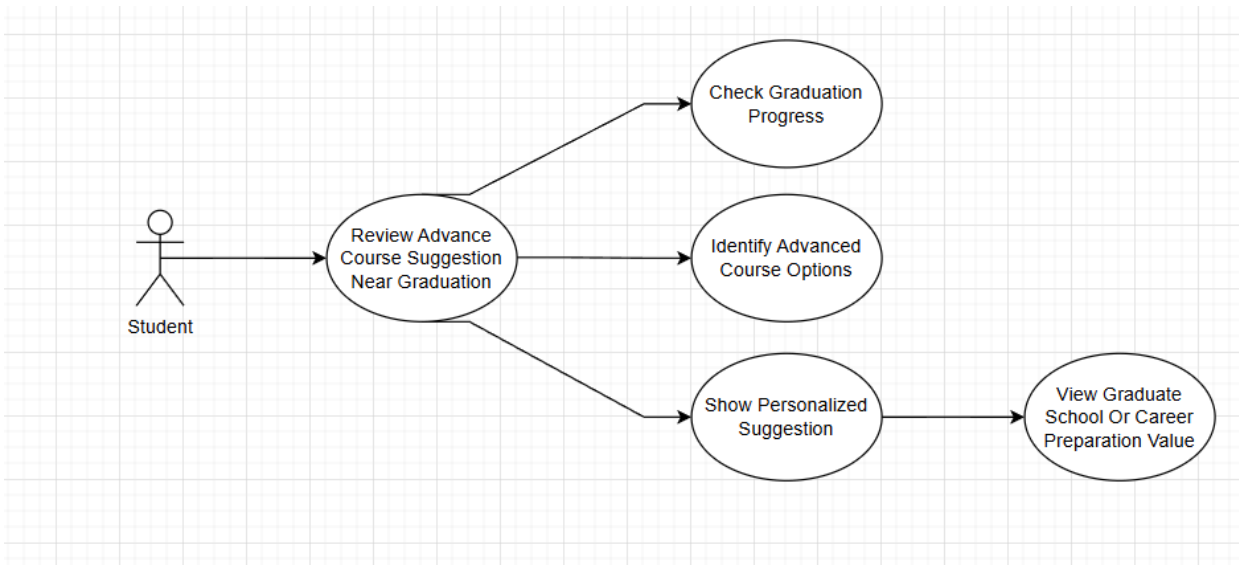
Use Case 4

- Actor
 - Student
- Assumptions
 - The student has degree progress information available.
 - Future course and requirement information exists in the system.
 - The student wants to plan beyond the current semester.
- Use Case
 - A student is trying to plan several semesters ahead and feels worried about delaying graduation. The student wants to know which courses should be taken later and how future classes may fit into their degree plan. The student uses Better Student Center to create a planned schedule for upcoming semesters.
 - The student reviews remaining requirements and adds possible future courses to a semester plan. This helps the student visualize their path toward graduation. The student feels more organized because they can see not only the current semester, but also future academic choices.
- Benefits for Student
 - Helps students make long-term academic plans.
 - Supports on-time graduation.
 - Makes future course planning easier to understand.
 - Gives students more confidence in their academic decisions.



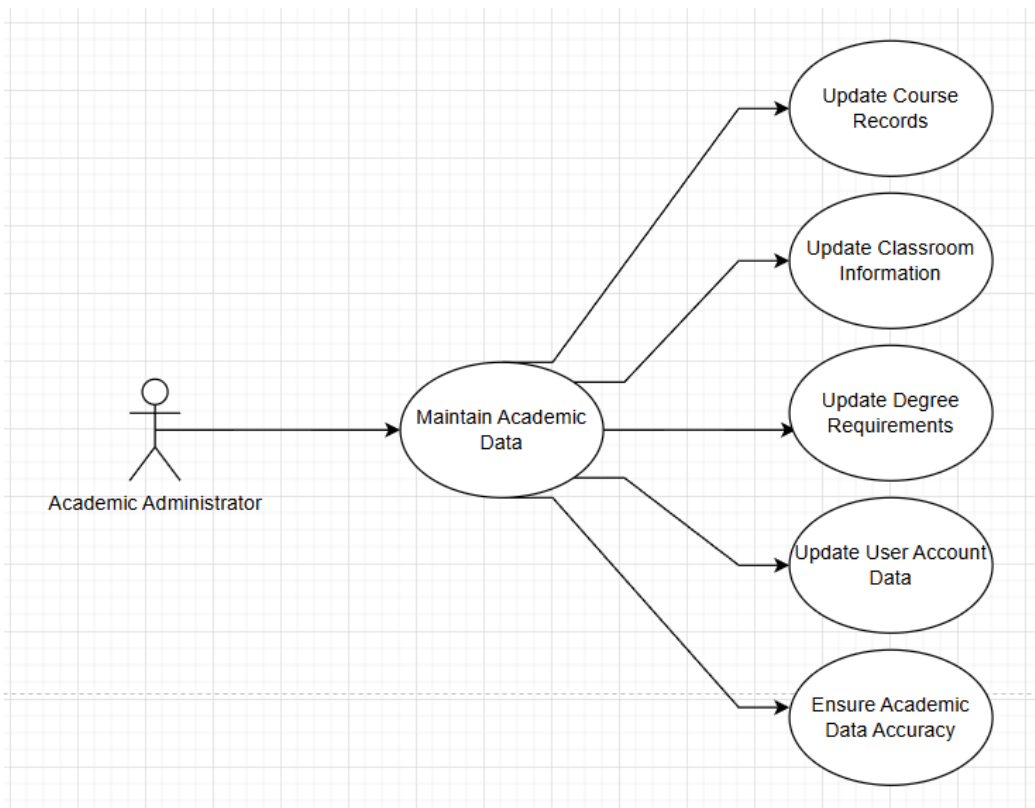
Use Case 5

- Actor
 - Student
- Assumptions
 - The student has degree progress data in the system,
 - The student is close to completing their degree.
 - Advanced course suggestion data exists in the system.
- Use Case
 - A student is close to graduation and starts thinking about graduate school, career preparation, or advanced academic interests. The student may not know which higher-level courses could help them prepare for the next step. Better Student Center reviews the student's academic progress and presents possible advanced course suggestions.
 - The student sees courses that may strengthen their knowledge, support graduate study preparation, or help with career goals. The student feels supported because the system does more than show minimum graduation requirements; it also helps them think about future academic development.
- Benefits for Student
 - Helps graduating students explore advanced academic options
 - Supports preparation for graduate school or career goals.
 - Encourages students to make stronger final semester course choices
 - Provides personalized academic guidance.



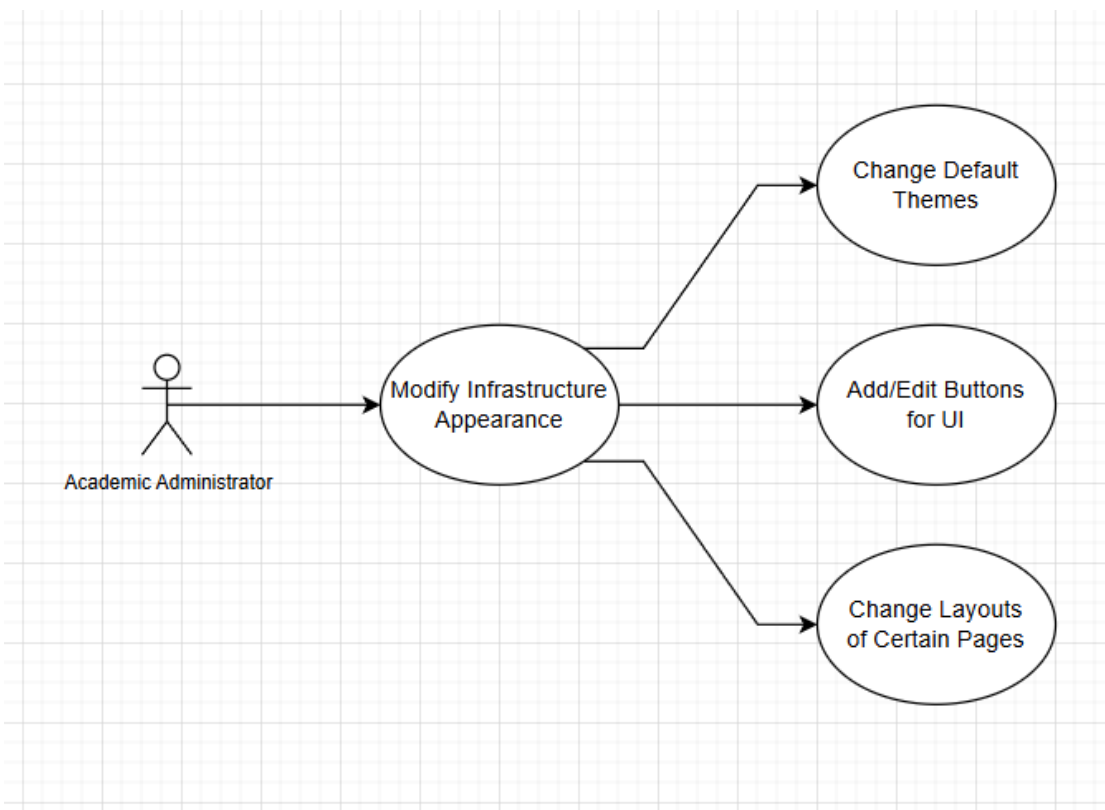
Use Case 6

- Actor
 - Academic Administrator
- Assumptions
 - The administrator has permission to manage academic data.
 - Course, classroom, degree requirement, and user data need to remain accurate.
- Use Case
 - An academic administrator is responsible for keeping university academic information accurate. Students and professors depend on correct course records, building information, classroom numbers, degree requirements, and account data. When changes occur, the administrator uses Better Student Center to update the relevant academic information.
 - By maintaining accurate records, the administrator helps ensure that students see correct schedules, professors see correct teaching assignments, and degree progress information remains reliable. The administrator feels confident that the system supports organized academic operations.
- Benefits for Academic Administrator
 - Keeps academic data accurate and consistent.
 - Supports reliable student and professor experiences.
 - Reduces errors in schedules, classrooms, and degree requirements.
 - Helps the university manage academic information in one centralized system.



Use Case 7

- Actor
 - Academic Administrator
- Assumptions
 - The administrator has permissions to change certain aspects of the infrastructure.
 - Different schools may have different needs or themes.
- Use Case
 - An administrator has permissions to manage certain properties of the UI layout, which includes the ability to modify, reorder, or recolor certain buttons, headers, or theme presets.
- Benefits for Academic Administrator
 - Access to superior customizability of the platform
 - Ability to make the platform feel more welcoming to students



Main Data Items and Entities

The Better Student Center application uses several main data items and entities to support student planning, enrollment, degree progress, campus navigation, and financial information.

User and Role Entities

- **User:** An individual who can access the Better Student Center system. A user may have different roles, such as Student, Instructor, or Admin, and the system displays features based on the user's assigned role.
- **User Account:** The login and authentication record for a user. It stores account information such as university login identity, role, account status, and contact information.
- **Student:** A user who can access student-centered academic tools, including course search, class schedule, degree progress, campus map, financial information, notifications, and course planning features.
- **Instructor:** A user who teaches assigned class sections and can view teaching schedules, classroom locations, and office hour information.
- **Admin:** A user with permission to manage academic data in the system, including courses, class sections, building locations, degree requirements, users, and student financial information.
- **Profile:** A user-facing representation of a student's or instructor's personal and academic information, such as name, university email, role, academic level, student type, declared major, and contact information.
- **Notification:** A system-generated alert that informs users about important updates, such as enrollment periods, approaching deadlines, schedule conflicts, insufficient walking time between classes, or successful and failed system actions.

Academic Entities

- **Department:** An academic unit that offers courses and majors. A department may have an office location, contact information, associated instructors, and a list of courses and majors.
- **Major:** A student's declared academic program. A major belongs to a department and contains degree requirements, required units, and catalog year information.
- **Degree Requirement:** A requirement that a student must complete for graduation. Degree requirements may include general education requirements, major required courses, major electives, unit requirements, and prerequisite rules.
- **Course:** An academic subject offered by a department. A course includes information such as course code, title, units, description, course level, course attributes, and prerequisites.
- **Class Section:** A specific offering of a course in a semester. A class section includes section number, instructor, meeting days and time, classroom, building, modality, enrollment status, seat availability, and waitlist information.
- **Semester:** An academic term in which class sections are offered. A semester includes term name, year, start date, end date, add/drop deadline, withdrawal deadline, and enrollment period.

- **Enrollment Information:** Information that tells a student when they are eligible to enroll, add, drop, or withdraw from class sections for a specific semester.
- **Enrollment Record:** A record of a student's academic history across semesters. It stores courses taken, semesters completed, enrollment status, and grades.

Schedule and Planning Entities

- **Planned Schedule:** A collection of class sections that a student is considering or planning to take. It stores selected class sections and helps detect time conflicts, missing prerequisites, and walking time between classes.
- **Calendar:** A weekly or monthly view of a student's planned and enrolled class sections. It displays course code, meeting time, instructor, building, and room information.
- **Cart:** A list of class sections selected by a student before final enrollment. The cart allows students to review selected sections, total units, enrollment status, and possible conflicts before enrolling.
- **Degree Progress Tree:** A visual academic roadmap that shows completed, planned, and remaining degree requirements. It helps students understand how their courses satisfy graduation requirements.
- **Course Details:** A detailed view of a course and its available class sections. It includes course description, prerequisites, section options, instructor information, schedule, location, and seat availability.

Campus Navigation Entities

- **Campus Map:** A map-based representation of campus buildings and resources. It allows students and instructors to locate classroom buildings, department buildings, and campus resources.
- **Building:** A campus structure where classes, departments, or resources may be located. A building includes a building name, location, and map coordinates.
- **Classroom:** A room inside a building where a class section may meet. A classroom includes room number, building association, and location information.

Financial Entities

- **Billing Profile:** A student's financial account profile in the system. It stores billing contact information and connects student charges, payments, refunds, and financial aid credits.
- **Student Charges:** Financial charges assigned to a student for a semester. Student charges include charge type, amount, due date, total outstanding balance, and payment status.
- **Financial Aid:** Aid information associated with a student, such as loans, awards, term, amount, and status. Financial aid may be applied toward outstanding student charges.
- **Transaction:** A financial activity related to a student's billing profile, such as a payment, refund, or financial aid credit.

Functional Requirements

1. User

- 1.1. A User shall be able to log in with their university's Single Sign-On (SSO)
- 1.2. A User shall be able to log out of their account
- 1.3. A User shall be identified as either a student or an admin
- 1.4. A user shall be shown features based on their assigned role.
- 1.5. A User shall be able to add or update contact information.

2. Student

- 2.1. A Student shall have an academic level, such as freshman, sophomore, junior, senior, or graduate.
- 2.2. A Student shall have a student type, such as first-time student, continuing student, transfer student, or international student.
- 2.3. A Student shall be able to view and update their profile.
- 2.4. A Student shall be able to search for courses.
- 2.5. A Student shall be able to filter courses.
- 2.6. A Student shall view course details.
- 2.7. A Student shall view enrollment information for the current term.
- 2.8. A Student shall be able to add to or remove courses from a planned schedule.
- 2.9. A Student shall be able to enroll in/add, drop, or withdraw from courses when enrollment is open for their student group.
- 2.10. A Student shall be directed to the appropriate Canvas course page when selecting the Canvas button for an enrolled course.
- 2.11. A Student shall view their planned schedule.
- 2.12. A Student shall view a Degree Progress Tree based on their declared major.
- 2.13. A Student shall be able to update their declared major for major planning purposes.
- 2.14. A Student shall view completed, planned, and remaining degree requirements.
- 2.15. A Student shall view missing prerequisites for planned courses.
- 2.16. A Student shall view class building locations on campus map.
- 2.17. A Student shall view outstanding charges.
- 2.18. A Student shall be redirected to the payment portal to make payments for outstanding charges.
- 2.19. A Student shall be redirected to the financial aid website for application.
- 2.20. A Student shall view their financial aid awards.
- 2.21. A Student shall be able to accept or decline their financial aid awards.
- 2.22. A Student shall view enrollment records from different semesters
- 2.23. A Student shall view and dismiss notifications.
- 2.24. A Student shall view their Cart
- 2.25. A Student shall be able to add or remove class sections from their Cart.

3. Admin

- 3.1.** An Admin shall be able to add, remove, or update course information.
- 3.2.** An Admin shall be able to add, remove, or update class section information.
- 3.3.** An Admin shall be able to add, remove, or update building location information.
- 3.4.** An Admin shall be able to add, remove, or update degree requirement information.
- 3.5.** An Admin shall be able to add, remove, or update student financial information.
- 3.6.** An Admin shall be able to modify student profile information.
- 3.7.** An Admin shall modify the home page layout.

4. Enrollment Information

- 4.1.** Enrollment Information shall display the date and time when the student's student type is eligible to begin enrollment.
- 4.2.** Enrollment Information shall display the enrollment window for the current term.
- 4.3.** Enrollment Information shall notify a student when enrollment is available for their student type.
- 4.4.** Enrollment Information shall display the add/drop deadline for the current term.
- 4.5.** Enrollment Information shall display the withdrawal deadline for the current term.

5. Course (Course Details)

- 5.1.** A Course shall have basic course information (code, title, units)
- 5.2.** A Course shall be associated with a department
- 5.3.** A Course shall have a detailed course description.
- 5.4.** A course shall have a course level (lower-division, upper-division, or graduate level)
- 5.5.** A course shall have course attributes (General Education areas, repeatable/nonrepeatable, lecture/lab)
- 5.6.** A Course shall have prerequisite information.
- 5.7.** A Course shall have available class sections for a selected term.
- 5.8.** A Course shall be searchable with the class search.
- 5.9.** A Course shall be filterable with the class search.

6. Class Section

- 6.1.** A Class Section shall have section details (section number, term, instructor name, meeting time, location, enrollment status, and seat availability).
- 6.2.** A Class Section shall be addable or removable from a student's planned schedule.
- 6.3.** A Class Section shall have a modality (in-person, online, hybrid)
- 6.4.** A Class Section shall have a meeting type (synchronous, asynchronous)
- 6.5.** A Class Section shall be addable to a student's cart
- 6.6.** A class section shall indicate whether it is eligible to be added to a cart.
- 6.7.** A Class Section shall indicate whether it conflicts with class sections already in a Student's planned schedule.

7. Planned Schedule

- 7.1.** A Planned Schedule shall store all class sections selected by a student.
- 7.2.** A Planned schedule shall display the relevant course information for each selected class section (code, section number, meeting time, location, modality, instructor, and enrollment status)
- 7.3.** A Planned Schedule shall allow a student to add or remove class sections.
- 7.4.** A Planned Schedule shall detect time conflicts between selected course sections.
- 7.5.** A Planned Schedule shall identify missing prerequisites for selected class sections.
- 7.6.** A Planned schedule shall calculate the walking time between two classes.
- 7.7.** A Planned schedule shall provide access to the Calendar view.

8. Campus Map

- 8.1.** A Campus Map shall display campus building locations.
- 8.2.** A Campus Map shall display the building location for a selected class section.
- 8.3.** A Campus Map shall display classroom location information for a selected class section.
- 8.4.** A Campus Map shall display campus resources (library, gym, student center, and health services).
- 8.5.** A Campus Map shall display the department building associated with a student's declared major.
- 8.6.** A Campus Map shall allow a student to return to the Calendar after viewing a class location.

9. Degree Progress Tree (Roadmap)

- 9.1.** A Degree Progress Tree shall display required General Education courses, Major electives, and Major requisites.
- 9.2.** A Degree Progress Tree shall display completed requirements, planned requirements, and remaining requirements.
- 9.3.** A Degree Progress Tree shall display remaining units needed for graduation.
- 9.4.** A Degree Progress Tree shall display prerequisite information for planned courses.
- 9.5.** A Degree Progress Tree shall recommend courses that satisfy remaining requirements.
- 9.6.** A Degree Progress Tree shall allow a student to view which courses can satisfy remaining requirements (GE, electives).
- 9.7.** A Degree Progress Tree shall update when a student updates their planned schedule.
- 9.8.** A Degree Progress Tree shall update when a student changes their declared major.
- 9.9.** A Degree Progress Tree shall display which requirement category a planned course satisfies.

10. Enrollment Record

- 10.1.** An Enrollment Record shall display a student's academic history across semesters.
- 10.2.** An Enrollment Record shall display the semester associated with each course
- 10.3.** An Enrollment Record shall display the grade associated with each course.

11. Student Charges

- 11.1.** Student Charges shall display charge information (charge type, amount, due date)
- 11.2.** Student Charges shall display the total outstanding balance.
- 11.3.** Student Charges shall display the estimated amount owed or refund amount after financial aid is applied.
- 11.4.** Student Charges shall provide a link to the university payment portal for payment processing.

12. Financial Aid

- 12.1.** Financial Aid shall display name, type (loan or award), term, amount, and status (pending/accepted/declined).
- 12.2.** Financial Aid shall be applied toward outstanding charges when applicable.
- 12.3.** Financial Aid shall provide a link or redirect to the university financial aid application website.

13. Semester

- 13.1.** A Semester shall show a term name, start date, end date, add/drop deadline, withdrawal deadline, and enrollment period.
- 13.2.** A Semester shall be able for users to enroll in the class sections.

14. Department

- 14.1.** A Department shall have a department name, office location, and contact information.
- 14.2.** A Department shall show a list of courses
- 14.3.** A Department shall list many majors.

15. Major

- 15.1.** A Major shall have a major name, department belong, required units, and catalog year.
- 15.2.** A Major shall show the degree requirements.

16. Degree Requirement

- 16.1.** A Degree Requirement shall have a requirement category, required units, and requirement status.
- 16.2.** A Degree Requirement shall be satisfied by courses.
- 16.3.** A Degree Requirement shall indicate whether the requirement is completed, planned, or remaining.

17. Billing Profile (Finances)

- 17.1.** A Billing Profile shall be provided to students.
- 17.2.** A Billing Profile shall store billing contact information and payment-related account information.
- 17.3.** A Billing Profile shall display all outstanding charges, payments, refunds, and financial aid credits.
- 17.4.** A Billing Profile shall calculate the current outstanding balance.

18. Student Profile

- 18.1.** A Student Profile shall display the user's student information (name, university, email address, role, academic level, student type, and declare major)
- 18.2.** A Student Profile shall allow a user to add or update their contact information.
- 18.3.** A Student Profile shall allow a student to update their declared major for degree planning purposes.

19. Notifications

- 19.1.** A Notification shall inform a student when their enrollment period begins.
- 19.2.** A Notification shall inform a student about approaching deadlines.
- 19.3.** A Notification shall warn a student about schedule conflicts, or insufficient walking time between classes.
- 19.4.** A Notification shall inform a student when an attempted action succeeds or fails.

20. Calendar

- 20.1.** A Calendar shall display a student's planned and enrolled class sections in a weekly or monthly format.
- 20.2.** A Calendar shall display each class section on the correct day and time
- 20.3.** A Calendar shall visually distinguish planned classes from enrolled classes.
- 20.4.** A Calendar shall display relevant course information (course code, meeting time, instructor, building and room)
- 20.5.** A Calendar shall allow a student to select a class section and view its course details.
- 20.6.** A Calendar shall allow a student to select a class location and open it on the Campus Map

21. Cart

- 21.1.** A Cart shall store class sections selected by a student.
- 21.2.** A Cart shall display relevant course information (Course name, number, units, instructor, meeting time, location, and enrollment status for each section)
- 21.3.** A Cart shall allow a student to add or remove class sections.
- 21.4.** A Cart shall calculate the total number of course units.

Nonfunctional Requirements

1. Performance

- 1.1. The system shall display a visual response within three seconds after a user submits a normal request under expected class-project usage.
- 1.2. The login process shall complete within five seconds after valid credentials are submitted.
- 1.3. The class schedule page shall load within three seconds under normal network conditions.
- 1.4. The campus map shall begin displaying within five seconds after a student selects a class location.
- 1.5. Degree progress calculations shall be displayed within five seconds after the student opens the Degree Progress page.
- 1.6. The system shall support at least 25 simultaneous users during prototype testing without becoming unavailable.

2. Reliability

- 2.1. The deployed application shall be available through a public URL during schedule demonstration and grading periods.
- 2.2. The system shall not lose previously saved schedule, degree progress, building, course, or office-hour data after a normal server restart.
- 2.3. The system shall display a clear error message when an operation cannot be completed.
- 2.4. A failed database operation shall not create incomplete or duplicate records.
- 2.5. The system shall confirm successful create, update, and delete operation visually to the user.
- 2.6. The PostgreSQL database shall be backed up at least once per week during active development.

3. Usability

- 3.1. A first-time student user shall be able to locate their schedule, campus map, degree progress, and planning pages within five minutes without formal training.
- 3.2. A student shall be able to open a class location from the schedule in no more than three user actions.
- 3.3. Navigation labels shall use clear terms such as Schedule, Campus Map, Degree Progress, Course Planning, and Profile.
- 3.4. Student, Professor, and Admin dashboard shall display only functions relevant to the user's role.
- 3.5. Required form fields shall be clearly marked, and validation errors shall identify the field that needs correction.
- 3.6. The interface shall remain readable at common laptop resolutions, including 1366 x 768 and 1920 x 1080.
- 3.7. The system shall provide basic responsive behavior for mobile screens with a width of at least 360 pixels.
- 3.8. Degree progress shall be separated visually into major requirements, major electives, general education requirements, and total credits.

4. Security

- 4.1. All public communication between the browser and the deployed applications shall use HTTPS.
- 4.2. The application shall use TLS 1.3 where supported by the client and server configuration.
- 4.3. User passwords shall be hashed using bcrypt before being stored in the database.
- 4.4. Plain-text passwords shall never be stored in the database or application log files.
- 4.5. The system shall require authentication before allowing access to Student, or Admin dashboards.
- 4.6. The system shall use role-based authorization to prevent students from accessing Admin functions.
- 4.7. Authentication tokens shall expire after a defined period of inactivity.
- 4.8. Database credentials, JWT secrets, and cloud credentials shall be stored in environment variables.

5. Privacy

- 5.1. The system shall collect only the information required for this project, including names, email address, user role, schedule information, completed courses, planned courses.
- 5.2. Student schedule and academic planning data shall be used only to provide schedule, map, degree progress, and course planning functions.

6. Compatibility

- 6.1. The application shall support the current version (149) of Google Chrome.
- 6.2. The frontend shall operate on Windows, macOS, and Linux through a supported web browser.
- 6.3. The frontend shall operate on other non-Windows platforms (maxOS, Linux) through a supported version of Google Chrome.
- 6.4. The system shall not require users to install browser plug-in.

7. Hardware and Networking Requirements

- 7.1. The application shall be deployed on an AWS Linux cloud instance.
- 7.2. The cloud instance shall have sufficient resources to run the React frontend, [Node.js/Express](#) backend, Nginx, and the required application services.
- 7.3. The application shall be accessible through a device with an internet connection and a supported web browser.
- 7.4. Remote database access shall be limited to authorized user and approved network addresses.

8. Media Content Requirements

- 8.1.** Team-member photos on the About page shall use JPEG, PNG.
- 8.2.** Each uploaded team-member photo shall not exceed 2 MB.
- 8.3.** Images shall include meaningful alternative text for accessibility.

9. Legal, Licensing, and External Requirement

- 9.1.** The system shall display a disclaimer stating that degree guidance and advanced course suggestions are planning aids and do not replace official academic advising.
- 9.2.** The system shall state that graduate or advanced course suggestions require confirmation under university and department policies.
- 9.3.** Course, degree, and policy information shown in the prototype shall be identified as sample data when it is not obtained from an official source.

10. Organizational and Development Constraints

- 10.1.** The frontend shall be developed using JavaScript and React.
- 10.2.** The backend shall be developed using JavaScript, [Node.js](#), and [Express.js](#).
- 10.3.** The system shall use PostgreSQL as its relational database management system.

11. Coding Standards

- 11.1.** Variables and functions shall be named and formatted according to the language's conventional syntax.
- 11.2.** API keys, and other private information shall not be stored inside a public folder.
- 11.3.** All code must need approval from at least one team member before being pushed to the main branch

Competitive Analysis

Competitor Information

Feature/C ompany	SFSU Student Center (Oracle Peoplesoft)	College of San Mateo (Ellucian Banner)	Infinite Campus	Ex Libris campusM	Workday Student
Strengths	It offers a comprehensive set of features, including everything a student might—or might not—need.	Their interface is very clean and does not present users with too much information at first glance.	Has features for students to check grades, assignments, announcements, and graduation reports. Can be used as a standalone classroom-esque application without a third party app like Canvas.	Well optimized mobile interface that has personalizable dashboards, campus map integration/class locations	Students can save schedules and register directly from their academic plans. Detailed class search filters and has schedule-conflict checks.
Weaknesses	The page contains too much information that is sparsely disorganized, and many features that are important to students are hidden or not immediately obvious.	The information provided on their website is too brief, so students may not know where they should go.	Basic implementation of necessary features. Reported to have security concerns	Primarily a mobile app. Web interface not as polished. Features such as registration, class searching, and degree progress depend on other applications.	Unintuitive, uninspiring user interface. Poor implementation of registration system.
Prices	License price: \$245 / FTE Student Software Update License & Support: \$53.90 / FTE Student / year, \$4.49 / FTE Student / month	Not publicly listed; institution-specific contract pricing. Public contract example: Portland Community College approved a 3-year Banner ERP licenses and support contract for \$2,592,481.	Tiered per-student pricing. \$6-20 per student annually. Cloud hosting costs an additional \$1 per student.	Not publicly listed.	Annual subscription price for Workday Student Service: \$166.38 / FTE (before implementation, integrations, consulting, training, or optional products)
Student Enrollment/Class Search	The Enrollment and Class Schedule is not together. Student has to go to other page to Scratch the class they want	On the main page, there is one option called "Register for Classes" and another called "Browse Classes." However, when students open one of these pages, they cannot access	Not available. Infinite Campus is primarily used by K-12 schools where education is continuous for students.	Class search and registration is done through a third party student information system rather than campusM itself.	Comprehensive class search system that directly links with saved schedule or academic plan system.

		the other page directly. As a result, students who want to enroll in classes have to constantly move back and forth between pages in order to browse and register for classes. This makes the process inconvenient and inefficient.			
Weekly Schedule	The Weekly Schedule will provide the information in a weekly calendar, But on the page it won't tell Student where is the Building Location is.	The weekly schedule is located inside the registration information section, which makes it hard for new students to find. Based on interview feedback, some students may not even know this feature exists unless someone tells them.	The weekly schedule is displayed through an extremely simple calendar.	Students can view their schedule by day, week, or year. Class detail is displayed, which can include the instructor, time, and location. Has support for the device's default map application also.	Class schedule is built into the registration and academic planning system. Has block-style course calendar view for students to see where each class occupies in their school day.
Degree Planner	The Degree Planner will provide the information to student, but on the Planner it won't show the Instructor of who is teaching the class for following semester, and if Student delete the course or want to make some change, the Student has to find the Class area by themselves; the System won't automatic provide the information to student	The system does not have a Degree Planner directly integrated into the Student Center. Instead, this function is placed on the school's main website as a separate resource. Because it is not connected to the Student Center, students may need to leave the student portal and search the school website manually to find degree planning information..	Infinite Campus has academic planner and graduation progress features. Students can plan courses across multiple school years and see whether they're meeting graduation requirements. (K-12 graduation requirements only)	Does not feature a built-in degree planner. Degree/schedule planning is done through a third party student information system rather than campusM itself.	Students can view completed, planned, and remaining course requirements and use their academic plan while preparing for registration.

Feature Comparison with Competitors

Feature	Oracle Peoplesoft	Ellucian Banner	Infinite Campus	Ex Libris campusM	Workday Student	Our Student Center
Campus Map Integration	-	-	-	++	-	+
Schedule Conflict Detection	-	+	+	-	+	++
Class Search/Enrollment	+	+	-	-	++	++
Class-Walking Time Calculation	-	-	-	-	-	+
Degree Progress Report	+	-	+	-	++	++
Weekly Schedule View	+	+	+	+	++	++

Competitive Analysis Summary & Validation

Our competitive analysis shows that some of our proposed features are not necessarily unique. Some features mentioned above already exist in some competing systems. However, the research validates our approach because our Better Student Center application implements and connects these features better than our competitors. For example, our system combines degree planner, class search, schedule conflicts, and walking distance all in one place instead of separating them across different pages or applications.

The class-walking time calculation is our most unique feature because none of the competitors check the distance between a student's classes. Therefore, our project differs from our competitors and is validated through better integration, clearer information, and just an overall more convenient student experience.

Project Readiness Checklist

The team (Big Backs) has established regular communication channels and a weekly meeting schedule. We meet in-person on Tuesdays and Thursday before/during class and the team lead hosts an asynchronous Discord meeting on Friday afternoon. An overview of the team meeting is posted after it concludes for team members who were not able to attend. Discord serves as the main platform for live discussion as well as ongoing communication.

Project responsibilities have been divided among team members, and the assigned roles are listed on the title page. Assigned roles may be changed during the course of the semester.

The team has finalized our technology stack and has started reviewing the tools needed for our application development. In addition, the repository has been organized to meet Milestone 1 requirements. Our technology stack is listed in the next section.

High Level Architecture and Technologies

Components in **bold text** are part of the instructor-approved tech stack.

Component	Configuration
Cloud provider	Amazon Web Services
SSL Certificate	AWS Certificate Manager
Application Server	AWS EC2
Operating System	Ubuntu Server
SSH Username	ubuntu
SSH Port	22
Database Service	Amazon RDS for PostgreSQL
Database Port	5432
Database Name	student_center
RDS endpoint	student-center-db.chwwk2s88tac.us-east-2.rds.amazonaws.com
Frontend	React with Vite
Backend	Node.js with Express.js
Web Server	Nginx
Repository Location on EC2	/home/ubuntu/csc648-848-su26-project-Big Backs-04
Public Frontend Directory	/var/www/html
SSL / HTTPS	HTTPS using TLS 1.3

Team Familiarity with Tech Stack

Team Familiarity Table

Team Member	Javascript	React	Node.js	PostgreSQL	AWS	TLS 1.3	average
Wen Chien Yen	3	3	3	2	3	0	2.33
Banghao Yuan	3	0	3	0	0	0	1.00
Sean Wang	3	1	3	2	0	0	1.50
Oscar Wang	4	0	3	1	0	0	1.33
Enoch Lin	3	1	3	2	0	0	1.50
Kashif Zada	2	1	3	1	0	0	1.17

We scored the major technologies that are central to our project stack. Smaller supporting libraries such as React Router, Axios, bcrypt, JWT, dotenv, Leaflet.js, and Chart.js are listed in the frameworks/libraries section, but they are not scored individually because they are supporting tools rather than the main technology stack. Our team will learn and apply these libraries as needed during implementation.

Overall Team Familiarity Summary

The team has the strongest familiarity with JavaScript and Node.js, which supports our decision to use JavaScript for both frontend and backend development. React familiarity is lower for some members, so frontend tasks may require additional learning and support. PostgreSQL familiarity is moderate to low, but it is still manageable for the project because the database design will focus on relational data such as users, schedules, courses, buildings, and degree requirements. AWS and TLS 1.3 familiarity are currently low, so deployment and security configuration may require extra preparation and documentation.

Team Contributions

Team lead assigned contribution scores. All team members have submitted confidential feedback via the instructor provided Google form.

Team Member	Wen Chien Yen	Oscar Wang	Banghao Yuan	Sean Wang	Enoch Lin	Kashif Zada
Contribution Score	7	6	6	6	4	4

Wen Chien Yen

For Milestone 1 Version 1, I worked on the first version of the use cases and nonfunctional requirements. For the use cases, I helped identify how the main actors, including Student, Professor, and Admin, would interact with the Better Student Center system. This included defining important user actions related to course planning, schedule viewing, campus map support, degree requirement guidance, and graduate planning suggestions.

I also contributed to the nonfunctional requirements by identifying quality expectations for the system, including usability, reliability, response time, privacy, compatibility, and security. These requirements helped explain how the system should perform beyond its basic features.

As Team Lead, I coordinated the team's project direction, task organization, and submission preparation. I helped clarify the project scope as the Better Student Center system and ensured that the team focused on the approved features. I also helped organize team responsibilities, communicate required changes, and review the documentation for consistency across features, requirements, data items, and infrastructure details.

For the technical setup and deployment preparation, I worked on AWS EC2, Nginx, GitHub Actions deployment, Amazon RDS PostgreSQL setup, credentials documentation, and instructor access verification. This helped prepare the project infrastructure for deployment, grading access, and future backend/database integration.

Sean Wang

For Milestone 1 Version 1, Sean contributed to the second version of the functional requirements. He also worked on the competitive analysis by writing the competitive analysis summary and completing approximately half of the competitor comparison information. In addition, he helped create the project readiness checklist, which supported the team in reviewing whether the project idea, scope, and documentation were prepared for submission.

For Milestone 1 Version 2, Sean revised the functional requirements and competitive analysis based on the instructor's feedback. His updates helped improve the clarity, organization, and accuracy of the requirements and competitor comparison sections.

Oscar Wang

For Milestone 1, Oscar helped with the early formatting of the project document and performed general grammar checks and revisions. He reviewed and improved the executive summary and unique features sections to make the project description clearer and easier to understand.

Oscar also contributed the first version of the main data items and entities. This helped the team begin identifying the important data objects needed for the Better Student Center system, including the information that would later support the database design and functional requirements.

Also for M1V2, added additional page break formatting to the document to be more edit-friendly, as well as added one more use case #6 and its diagram for administrators.

Replaced the team alias for the repository.

Banghao Yuan

For Milestone 1 Version 1, Banghao created the first version of the functional requirements. He also contributed approximately three-fifths of the competitive analysis and worked on the use case diagram. His work helped establish the initial system behavior, user interactions, and comparison with existing student center platforms.

For Milestone 1 Version 2, Banghao reworked the use cases and revised the use case diagram based on the instructor's feedback. He also changed the format of the main data items and entities section and added missing information identified in the feedback. These revisions helped make the documentation more complete and better aligned with the project requirements.

Enoch Lin

For Milestone 1, Enoch contributed to the use case section by helping with some of the use case content and reviewing how users would interact with the Better Student Center system. His work supported the team's early planning of student and administrator interactions.

Enoch also completed his individual part of the throw-away prototype by adding and pushing his own photo and personal introduction to the public About page. This helped complete the deployed team webpage requirement and ensured that his profile information was visible on the public project URL.

Kashif Zada

For Milestone 1, Kashif contributed to the competitive analysis section by helping with part of the competitor research and comparison. His work supported the team's analysis of existing student center platforms and helped identify how the Better Student Center project could provide value through its proposed features.

Kashif also completed his individual part of the throw-away prototype by adding and pushing his own photo and personal introduction to the public About page. This helped complete the deployed team webpage requirement and ensured that his profile information was visible on the public project URL.